RAJALAKSHMI ENGINEERING COLLEGE

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M2

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|---|---|
| **Started on** | Saturday, 3 October 2026, 12:58 PM |
| **State** | Finished |
| **Completed on** | Saturday, 3 October 2026, 1:23 PM |
| **Time taken** | 24 mins 43 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int slow = arr[0];
    int fast = arr[0];


    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);


    slow = arr[0];

    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 5<br>1 2 3 4 4 | 4 | 4 | ✓ |
| ✓ | 5<br>1 1 2 3 4 | 1 | 1 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course